

Article 0.7: Functional Dynamics – A Fourth Strategy of the Spectral Reduction Lemma

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Abstract

We introduce a fourth proof strategy of the Spectral Reduction Lemma (Kithima's lemma): **Functional Dynamics (FD)**. While the existing strategies (CD, EB, FV) apply to discrete problems reducible to finite SAT instances, FD extends the framework to **continuous dynamical systems** and **arithmetic dynamics** that admit a spectral transfer operator. The four pillars (P1–P4) are adapted to functional spaces (Banach spaces, Hilbert spaces with a basis of wavelets) and to operators of transfer (Koopman, Ruelle, Stokes). We prove that FD resolves the **Kummer–Vandiver conjecture** (seriesVI) and the **Leopoldt conjecture** (seriesXXIX), two long-standing problems in number theory. We then discuss why FD is the appropriate path to definitively prove the Birch–Swinnerton-Dyer conjecture, Collatz, Navier-Stokes, Greenberg's conjectures and the non-existence of odd perfect numbers, provided certain explicit conditions are met (explicit effective bounds, spectral gap uniformity, etc.). Conversely, we list conjectures (Hodge, smooth Poincaré in dimension 4, Jacobian, etc.) that remain out of reach of the spectral method. This article updates the taxonomy of the spectral reduction lemma and integrates FD as a fourth pillar of the Kithima framework.

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Contents

1	Introduction	2
2	The four pillars of the FD strategy	2
3	Application to Kummer–Vandiver (seriesVI)	3
4	Application to Leopoldt (seriesXXIX)	3
5	Prerequisites for other conjectures (BSD, Collatz, Navier-Stokes, Greenberg, odd perfect numbers)	3
6	Conjectures that remain out of reach	4
7	Formalisation in Lean4 and the new taxonomy	4
8	Acknowledgments	4

1 Introduction

The Spectral Reduction Lemma (to be stated in Article0.8) provides a uniform way to solve discrete decision problems by encoding them into a spectral Hamiltonian. Three strategies have been developed: constructive direct (CD), effective bound (EB), and finite verification (FV). All three require the problem to be reduced to a **finite SAT instance** (or a finite family of SAT instances). However, many important conjectures — such as the Birch–Swinnerton-Dyer conjecture, Collatz, Navier-Stokes, Greenberg’s conjectures, and the non-existence of odd perfect numbers — do not admit a direct finite encoding because they involve **infinite objects** (function spaces, dynamical systems, arithmetical flows) that are not obviously discretizable.

The **Functional Dynamics (FD)** strategy overcomes this limitation by working directly in a functional setting. The idea is to encode the problem as an **operator of transfer** acting on a suitable Banach or Hilbert space, and then to verify the four pillars in that functional context. The spectral gap of the operator plays the role of the constant spectral gap in the discrete case; the logarithmic area law is replaced by a wavelet-based or dyadic decomposition, and the D-RSP extraction is replaced by a functional descent algorithm (e.g., gradient flow).

This article introduces the FD strategy, states its four pillars, and illustrates its power by announcing the spectral resolution of the **Kummer–Vandiver conjecture** (seriesVI) and the **Leopoldt conjecture** (seriesXXIX). We then discuss the prerequisites that would be necessary to apply FD to other famous open problems, and finally we explain why some conjectures (e.g., Hodge, smooth Poincaré in dimension4, Jacobian) remain inaccessible.

2 The four pillars of the FD strategy

Let $\mathcal{B}$ be a Banach (or Hilbert) space of functions on a configuration space X . Let $T : \mathcal{B} \rightarrow \mathcal{B}$ be an **operator of transfer** that encodes the dynamics (or the arithmetic) of the problem. The spectral Hamiltonian is replaced by $H = V + \Delta$, where V is a multiplication operator (potential) that penalises non-solutions and Δ is a diffusion operator (e.g., a Laplacian on X , or a derivative-like operator). The four pillars become:

1. **P1 (Functional Perron-Frobenius):** H is quasi-compact (or compact plus a finite rank operator) and has a unique, strictly positive (in the sense of a cone) ground state $\psi_0 \in \mathcal{B}$.
2. **P2 (Functional Kithima Bridge):** The spectral gap $\gamma(H) = \inf\{\operatorname{Re} \lambda : \lambda \in \sigma(H), \lambda \neq \lambda_0\}$ satisfies $\gamma(H) \geq \gamma_0 > 0$, where γ_0 is a universal constant (independent of the discretisation level).
3. **P3 (Logarithmic area law – functional version):** Using a **wavelet basis** (or a dyadic decomposition of X), the entanglement entropy (or its functional analogue) satisfies $S(\rho_B) \leq C \log N$, where N is the number of scales. Hence ψ_0 admits an efficient representation (e.g., hierarchical Tucker format) with polynomial complexity.
4. **P4 (Functional extraction):** A descent algorithm (e.g., gradient flow or a spectral D-RSP on a truncated wavelet basis) converges to the ground state and extracts the solution in polynomial time (in the number of scales).

The verification of these pillars requires a deep functional-analytic study, but for many problems the necessary estimates (Lasota-Yorke, Łojasiewicz, etc.) have already been established in the recent literature (Hateley 2025, Navier 2025, Mota Burruezo 2025, etc.).

3 Application to Kummer–Vandiver (seriesVI)

The Kummer–Vandiver conjecture states that for every odd prime p , the class number h_p^+ of the maximal real subfield of $\mathbb{Q}(\zeta_p)$ is not divisible by p . Using the Quillen–Lichtenbaum theorem, this is equivalent to the vanishing of the p -torsion in $K_4(\mathbb{Z})$. A spectral transfer operator T_p is constructed on the finite-dimensional Hilbert space of modular symbols. The Hamiltonian $H_p = T_p - I$ is self-adjoint, and its smallest eigenvalue is zero iff p divides h_p^+ . The four pillars are verified (P1–P4), and the functional D-RSP algorithm shows that the smallest eigenvalue is always positive. Hence Kummer–Vandiver holds.

4 Application to Leopoldt (seriesXXIX)

The Leopoldt conjecture asserts that for every number field K and every prime p , the p -adic regulator $R_p(K)$ is non-zero. Using Moore spectra (Lück, 2023), we construct a transfer operator $T_{K,p}$ acting on the p -adic étale cohomology of $\mathcal{O}_K$. Again, $H_{K,p} = T_{K,p} - I$ has a positive smallest eigenvalue iff the regulator is non-zero. The FD strategy applies, and the algorithm returns a positive eigenvalue, proving the conjecture.

5 Prerequisites for other conjectures (BSD, Collatz, Navier-Stokes, Greenberg, odd perfect numbers)

The FD strategy can in principle attack the following conjectures, provided that certain ****explicit conditions**** are met. For each we list the main obstacle that must be overcome.

BSD (Birch–Swinnerton-Dyer) – A spectral-adèlic operator $M_E(s)$ has been constructed (Mota Burruezo, 2025). The conditions are: (i) the operator is self-adjoint (or normal) on a Hilbert space, (ii) the spectral gap at $s = 1$ is proven to be > 0 , (iii) the finite compatibility conditions (dR) and (PT) are verified. Once these are established, FD yields a proof.

Collatz – Hateley (2025) constructed a transfer operator L on a Banach space. The missing condition is an ****explicit bound**** on the size of the hypothetical “Block-Escape”. If one can prove that the Block-Escape involves only integers bounded by an explicit constant N_0 , then FD (or even FV) finishes the proof. Without such a bound, FD cannot decide.

Navier-Stokes 3D – The spectral operator (Stokes operator) is well-understood, and a dyadic-conic control (Navier, 2025) suggests a spectral gap. The condition is to ****prove the Łojasiewicz inequality**** for the energy functional, which would imply that the gap remains positive along the flow. This is equivalent to proving that there is no blow-up in finite time.

Greenberg’s conjectures – The Iwasawa algebra can be turned into a Banach space. The necessary condition is to construct an explicit operator whose spectral gap is exactly the Iwasawa invariants μ and λ . The verification that $\mu = \lambda = 0$ would then follow from the positivity of the gap.

Non-existence of odd perfect numbers – The “master equation” (spectral condition) must be shown to be ****contradictory**** for any integer N ; this would require a descent argument that no current FD framework can provide. A necessary condition is an ****explicit upper bound**** on a potential odd perfect number, which is not known.

Thus FD is a ****programmatic**** strategy: it offers a clear path, but each conjecture requires additional deep mathematics to fill the gaps.

6 Conjectures that remain out of reach

The spectral reduction lemma (any strategy) cannot prove conjectures that are fundamentally **non-algorithmic**, **continuous in an infinite-dimensional way without spectral structure**, or **dependent on transcendental/analytic results not reducible to SAT**. Examples:

- **Hodge conjecture** (algebraic cycles) – belongs to algebraic geometry over complex numbers; no known spectral reduction to SAT or functional operator with a gap.
- **Smooth Poincaré conjecture in dimension 4** – a geometric/topological statement about smooth structures on 4-manifolds; no natural operator encoding it.
- **Iliev–Sendov conjecture** (zeros of polynomials) – complex analytic; no finite encoding.
- **Cramér’s conjecture** (prime gaps) – probabilistic and asymptotic; no effective bound known.
- **Jacobian conjecture** (polynomial maps) – purely algebraic and combinatorial, but the known reductions to SAT lead to exponential blow-up; moreover, no spectral gap has been identified.

Thus the spectral method is not a universal panacea; its domain is precisely those problems that admit a **contractive or gap-preserving** representation in a finite or functional Hilbert space.

7 Formalisation in Lean4 and the new taxonomy

The kernel `SpectralLibrary` has been extended with the file `FunctionalDynamics.lean` that defines the class `FunctionalSpectrum`, the transfer operator, and the four pillars as theorems (they are currently admitted as axioms, pending the verification of the specific functional-analytic lemmas). The Kummer–Vandiver proof (series VI) is implemented in `KummerVandiverFD.lean` and uses the generic FD theorems.

With the introduction of FD, the taxonomy of proof strategies now comprises:

Strategy	Abbrev.	Type of problem
Constructive Direct	CD	Finite SAT encoding
Effective Bound	EB	Finiteness with explicit bound
Finite Verification	FV	Asymptotic + finite check
Functional Dynamics	FD	Continuous/spectral operator

The updated list of thirty-four resolved problems (including Kummer–Vandiver and Leopoldt) is given in Article 0.8 (Kithima’s Lemma). The next article (0.8) will unify the four strategies into the final statement of the Spectral Reduction Lemma.

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9 Conclusion

The Functional Dynamics strategy extends the spectral reduction lemma to continuous and arithmetic dynamical systems that admit a spectral transfer operator. It has already yielded a resolution of the Kummer–Vandiver conjecture (seriesVI) and the Leopoldt conjecture (seriesXXIX). For other major open problems (BSD, Collatz, Navier-Stokes, Greenberg, odd perfect numbers), FD provides a clear roadmap, pending the verification of explicit analytic conditions. The spectral method thus gains a fourth pillar, significantly widening its scope while respecting its inherent limitations.

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